

# Building a unified science of sensorimotor control with Densely Observable Measurement Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

## 1. Mission

The MISSION of the **FreeMoCap Foundation X-Lab [FMC-X]** is to create a novel research apparatus called a **Dense Observation Measuring Environment [DOME]** that fuses heterogeneous instrumentation for the measurement or manipulation of the *perceptual, motor, and environmental* factors relating to the Agent/Environment interaction into a single convergent stream of calibrated, synchronized, semantically-coherent measurements through validated, uncertainty-traced, hardware-agnostic, sensor-to-model pipelines developed and disseminated in collaboration with a broad network of cross-disciplinary researchers to build a unified science of the sensory/motor control of biological and artificial agents in real-world environments.

### 1.1. Dense Observational Measuring Environment

- **DOME-L:** Flagship in Greater Boston area. Large enough to fully encompass smaller DOME-S and DOME-Mobile units to assist development and validation, and stress test ability to calibrate large spaces.
- **DOME-S:** Extension of the USB-camera based motion capture volumes we have been building as part of the FreeMoCap Project. Representative of the kinds of DOME that would be built in a research lab or classroom.
- **DOME-Mobile:** Wearable, mobile DOME, capable of generating the precisely the same model Dense Observations as a static DOME in unconstrained indoor or outdoor environments. Continues the lineage of PI's Gaze/Gait and Retinal Optic Flow research<sup>1-3</sup>

### 1.2. Targets

- **Measurement target** - Veridical estimates of retinal projection during real-world locomotion<sup>2-4</sup>
- **Integration goal** - Full pathway mapping from photon-retinal transduction on one end, and reaction force detection/generation at the substrate boundary (i.e. the ground)
- ▶ Buy what we can, build what we can't
  - ▶ Build as much of the whole thing with off-the-shelf equipment as we can, then build what doesn't exist
  - ▶ Let the research questions, research-bottle necking friction, and uncertainty-tracing drive development.
  - ▶ **Technical Bottlenecks that need solved** [TODO - re-org this into a more coherent 'Problem -> Solution' format]
    - **Eye trackers** - No open mobile eye trackers exist. Pupil still makes the Core, but its dev is frozen. No modern eye tracker can measure torsion or lens accommodation at any price. We can build our own eye trackers, but in the mean time we still use the best available (Pupil Core) and partially hydrate model with the best data we have.
      - After we make our cool new eye tracker, we use the newly unlocked DoF along with from previously existing DoFs to train a model that lets you backfill old data with estimations of previously unmeasured DoF.

- **Centrally Controllable Programmable Capture Volume** - Moving cameras around is a bottlenecking-friction (limits research because it's too hard to move the cameras), so we build a linked array of centrally and programmatically controllable camera mounts that control extrinsics and intrinsics, to create a system for a self-deploying capture volume (select region in available volume and cameras configure themselves for maximal coverage)
- **Hybrid Camera/IMU Kinematics** - Outside-in mocap systems (e.g. camera-based) produce high-accuracy/low-precision kinematic estimates (noise estimate centered on True value), Inside-Out systems (e.g. IMU-suits) produce high-precision/low-accuracy estimates. Their hybrid should be much better than either alone.
  - we will use the DOME-Mobile nested in the DOME-W to develop and validate this sensor-fused hybrid estimate. This way, we can create full interoperability between Human estimates derived from cameras, IMU's or both
- **Drone-Swarm Mocap** - Drone swarm to create hybrid mocap during outdoor locomotion

### 1.3. Data consumers

- Train better models using reprojected 3d estimates onto each camera's estimates
- Create validation dataset for consumer labs
- Train new models using new data of extreme movements (fix COCO datasets are old/limited problem)
- Package DOM datasets for RL training in robots, use resultant control theoretically models as testable hypotheses [Cite that thing that says training NN on a task predicts neuron firing for animal performing that task]

### 1.4. Environment Manipulation

- Animals - Like the BS magnet toy gantry
- Floor panels
  - 3d terrain - For control, and validating egocentric mapping from head-sensor array
  - Soft, slippery, sticky substrate
  - LED floor and/or Wall Panels
    - for ground and walls (Like a CAVE, ARGPv3<sup>5</sup> matthis2014 matthis2015 matthis2017)
    - Stimulus manipulation and gameification tasks (bug squishing, stepping stone, obstacle avoidance, dual-attention, etc)
- VR/Projector - Build equivalent methods in LED, VR, and Projector methods - measure uncertainty at those simulation boundaries (i.e. people step weird in VR, can we figure out why using VR + LED comparison?)

### 1.5. Vision

The perception/action loop of biological and artificial agents follows the same pattern: sample a thin slice of the available environmental energy through a limited set of imperfect transducers, then generate reaction forces against a substrate in service of some goal. Information flows in, forces flow out; the brain exists to yank the bones around.

Many disciplines within the broad vehicle of science point their lenses at key components of this complex exchange sensation, perception, decision, and action. Perceptual neuroscientists built magnificently precise tools and theories to measuring and explicating the neural mechanisms behind the cascade of neural activity which follows the absorption of a photon by an opsin molecule in the retina. Musculoskeletal biomechanists have characterized the stretch/

strain curves the winding titan filaments in our muscle spindles and mapped them to the full-body muscle activity of human movement [TODO - rewrite so its F&B reaching - photon - retina - V1 - etc, GRF - GTO - Muscle, motor - spine - muscle - force, or something].

Each member of these individual research clades understands that their work is contextualized in a broad and deeply cross-disciplinary domain of inquiry, and each strives to reach out across those disciplinary divides to their allies in nearby fields. But in practice, most of those threads fail to connect. Walk through the poster session of a visual neuroscience conference, you will mostly encounter variations of two-alternative-forced-choice-center-out-fixation tasks in non-mobile, head-fixed animal models. At the motor neuroscience conference, it will be single-joint planar reaching movements through a force field. At the biomechanics conference, its steady state walking on a treadmill. Each clade builds beautifully nuanced and complex theories in their domain of inquiry, but the pieces never quite fit together to form a coherent whole. Each group occupies a different room in the tower of Babel. They learned the nuances of different instruments, mastered different branches of mathematics. They are motivated by different kinds of arguments and ground their assertions in different flavors of statistics. They call out to each other in the first and last paragraphs of their research articles, the distant call of an unmet ally echoing across the yawning darkness between the fields.

And yet we are a species of explorers and we are drawn to those dark spaces. Each new generation of scientist builds new tools for their exploration; most dig down within their specialized niche, but some brave few build outwards towards the distant glowing shapes in the fog. Some of those even succeed, building new intergrations of previously disjointed theory - shedding light on new territory and the promise of new science at the intersections between deep domains of existing knowledge. But the discoverer cannot stay there. As reward for their success, they are bundled up and shipped to a new lab across the world with a rubber stamp on their forehead that reads "THIS ONE KNOWS NEUROSCIENCE."

Except they don't really, do they? Their expertise is in Neuroscience **and** Biomechanics **and** Robotics **and** the a few other disparate skillsets as well. They know more about each of those topics than *almost* anyone in the world... except for the all the people who will be evaluating their progress, scoring their grants, and reviewing their papers. To a specialist, cross-disciplinary research

Mentorship-based academic research has it backwards - Tools should be built by masters and taught to students, not the other way around. We should bake the wisom of our experise and long experience into the tools put into the hands of students and other researchers. Most users of the tool will never notice the myriad ways whey have been gently guided away from pinch points and spinning blades of empirical research, and the ones that do may become the next generation of tool builders.

We must separate the building of tools from the usage of them,

blah blah blah conclusion

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Figure 1: Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah  
blah captions. Blah blah blah captions.

**2. Technology Landscape**

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practitioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehavioral Data into those simulation/RL training environments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - <sup>1-3</sup>
  - etc
  - Ferret work
  -

## 3. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperability across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via composable data sources and associated downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

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| 3.1. Timeline |           |                                                                                                                                                                                                                                                                                                                                     |
|---------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase         | Name      | Description                                                                                                                                                                                                                                                                                                                         |
| 1             | Organize  | Set up Organization (Phase 0) – Solidify release of FreeMoCap v2                                                                                                                                                                                                                                                                    |
| 2             | Architect | Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)                  |
| 3             | Execute   | Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system                                                                                                                                                                                        |
| 4             | Stabilize | Establish OODA loop within expected standard operations                                                                                                                                                                                                                                                                             |
| 5             | Expand    | Once stabilized, expand key performance metrics                                                                                                                                                                                                                                                                                     |
| 6             | Align     | Focus on building deeply integrated community of users and contributors (Transition BDFL → BDFaSPoT)                                                                                                                                                                                                                                |
| 7             | Establish | Focus on long term stability and healthy growth                                                                                                                                                                                                                                                                                     |
| 8             | Maintain  | Beyond proposal timeline – Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending ‘mesoscale research revolution’ to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus) |

Table 1: Timeline of Phased Milestones

4. Senior/Key Personnel Qualifications

4.1. Jonathan Matthis, PhD [Principle Investigator]

4.2. EI

4.3. AC, PhD

4.4. RR, CPA or w/e

| Name / Role                            | Qualifications                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jonathan Matthis, PhD<br>President/CEO | <ul style="list-style-type: none"><li>• Founder and chief maintainer of FreeMoCap project</li><li>• Left tenure track academic research position precisely because the institution could not support the work that needed to be done</li><li>• Decades experience integrating visual neuroscience with biomechanics/robotics<sup>1-3</sup></li></ul>                                                                                      |
| EI<br>CTO                              | <ul style="list-style-type: none"><li>• Decades of experience in software Industry</li><li>• Chief architect and CTO for established companies</li><li>• Represents expertise truly absent from academic contexts</li><li>• Ensures world-class enterprise-level software capability</li><li>• Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science</li></ul> |
| AC, PhD<br>CSO                         | <ul style="list-style-type: none"><li>• Expertise in validation of clinical research tools and bench-to-bedside development</li><li>• [CITE DISSERTATION]</li><li>• Experience in neuroprosthetics design</li></ul>                                                                                                                                                                                                                       |
| RR<br>CFO                              | <ul style="list-style-type: none"><li>• Bookkeeping</li><li>• NSF Grant management</li><li>• Finances</li><li>• Entrepreneurship</li></ul>                                                                                                                                                                                                                                                                                                |

Table 2: Senior/Key Personnel Qualifications

## 5. Team Capabilities Statement

### 5.1. Senior Team Capabilities

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

### 5.2. Collaborators

| Initials           | Model / Domain       | Expertise                                  |
|--------------------|----------------------|--------------------------------------------|
| BS                 | Ferrets              | Visual/Perceptual neuroscience, ephys      |
| MD                 | Guinea Fowl          | Musculoskeletal Biomechanics, muscle units |
| DF                 | Mice                 | System Biology                             |
| AH                 | Marmosets / NHP      | Ephys                                      |
| MH                 | Human                | Vision and CPS                             |
| Mayo Clinic        | Human                | Brain scans and implanted sensors          |
| BD, GN/RAI, CH, JH | Robots / Fabrication | Control theory                             |
| AA                 | Physical facilities  | Rapid iteration, recruiting                |

Table 3: Collaborator Network

### 5.3. Governance

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosendaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

### 5.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

**Bibliography**

1. Matthis, J. S., Yates, J. L. & Hayhoe, M. M. Gaze and the Control of Foot Placement When Walking in Natural Terrain. *Current Biology* **28**, 1224–1233.e5 (2018).
2. Matthis, J. S., Muller, K. S., Bonnen, K. L. & Hayhoe, M. M. Retinal Optic Flow during Natural Locomotion. *PLOS Computational Biology* **18**, e1009575 (2022).
3. Muller, K. S. *et al.* Retinal Motion Statistics during Natural Locomotion. *eLife* **12**, e82410 (2023).
4. Muller, K. S. *et al.* Foothold Selection during Locomotion in Uneven Terrain: Results from the Integration of Eye Tracking, Motion Capture, and Photogrammetry. <https://elifesciences.org/reviewed-preprints/91243v2> (2024) doi:10.7554/eLife.91243.2.
5. Matthis, J. S. & Fajen, B. R. Humans Exploit the Biomechanics of Bipedal Gait during Visually Guided Walking over Complex Terrain. *Proceedings of the Royal Society B: Biological Sciences* **280**, 20130700 (2013).